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1. Abstract

This study presents the design and implementation of a dual-axis solar tracking system that determines the solar azimuth and elevation angle using differential signals from four LDR sensors. The system consists of two independent units: an STM32G031K8 microcontroller for motor control and an ESP32 microcontroller for power measurement and data logging. Power monitoring is performed via an INA226 module over the I2C protocol.

The control architecture employs a multi-layer signal processing pipeline consisting of a Kalman filter, moving average, a full PID controller with anti-windup and derivative filter, hysteresis dead band, and exponential position smoothing. Power data acquired across a $47\,\Omega$ dummy load is monitored in real time through a live web dashboard running on the ESP32 and recorded in CSV format.

Comparative A/B testing between the tracking and fixed panel configurations shows that the tracker produces %147.5 more gross energy over a three-hour measurement session. A parasitic consumption analysis reveals that the MG996R servo motors present a disproportionate load relative to the panel capacity, highlighting the critical role of actuator selection in small-scale tracking systems.

2. Introduction

Panel efficiency in solar energy systems is directly related to the angular orientation of the panel relative to incident irradiance. A fixed-mount photovoltaic panel can only achieve its optimum angle for a limited period each day due to the apparent motion of the sun. Solar tracking systems aim to minimise this loss by continuously orienting the panel perpendicular to the incoming radiation [1].

While single-axis trackers follow only the azimuth angle, dual-axis systems control both azimuth and elevation simultaneously to achieve higher energy yield [2]. This work presents a system in which a solar panel mounted on a two-servo pan-tilt mechanism is directed by differential signals derived from four LDR sensors.

The primary objectives of the project are:

- Sun angle estimation using four cross-placed LDR sensors
- Jitter-free, precise servo control through a multi-layer control architecture
- High-accuracy power and energy measurement with the INA226
- A live ESP32-based web dashboard with browser-side CSV recording
- Quantitative efficiency comparison against a fixed panel

3. System Architecture

The system is intentionally split into two isolated units to ensure that data acquisition is never affected by motor control timing or interrupt latency.

Tracker Unit (STM32): Processes LDR signals and drives the servo motors via PWM. All control algorithms run on this unit.

Logger Unit (ESP32): Reads panel power data from the INA226, serves a live web dashboard, and handles CSV recording. It has no direct connection to the motor control unit.

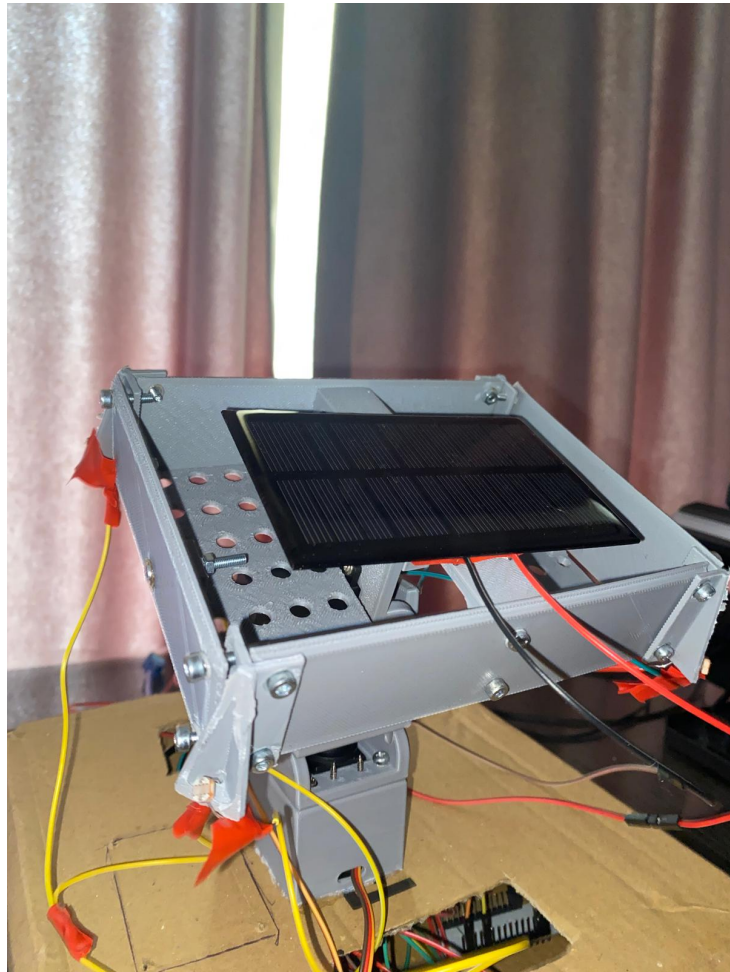


Figure 3.1: Assembled dual-axis solar tracking system

4. Hardware and Circuit Design

4.1 Bill of Materials

Table 4.1: Project bill of materials

Component	Role / Specification	Qty
STM32-NUCLEO-G031K8	Primary MCU — motor and LDR control	1
ESP-WROOM-32	Secondary MCU — logging and web server	1
LDR	Light sensor, 10 k Ω voltage divider	4
INA226	Bi-directional power monitor (I2C, 0.1 Ω shunt)	1
Dummy Load	47 Ω 5 W ceramic resistor	1
Servo Motor	Pan and tilt axis actuators	2
Solar Panel	6 V 150 mA monocrystalline	2
Resistor	10 k Ω LDR voltage dividers	4
Power Supply	5 V 3 A external adapter	1
Mechanical	Custom 3D-printed PLA pan-tilt structure [3]	—

4.2 Circuit Design

A critical design decision is powering the servo motors from an external 5 V adapter rather than the STM32's 3.3 V rail. The inrush current drawn by the servos would otherwise cause voltage drops on the MCU power rail, corrupting ADC readings. The 47 Ω dummy load was selected to force the 6 V/150 mA panel to operate near its maximum power point.

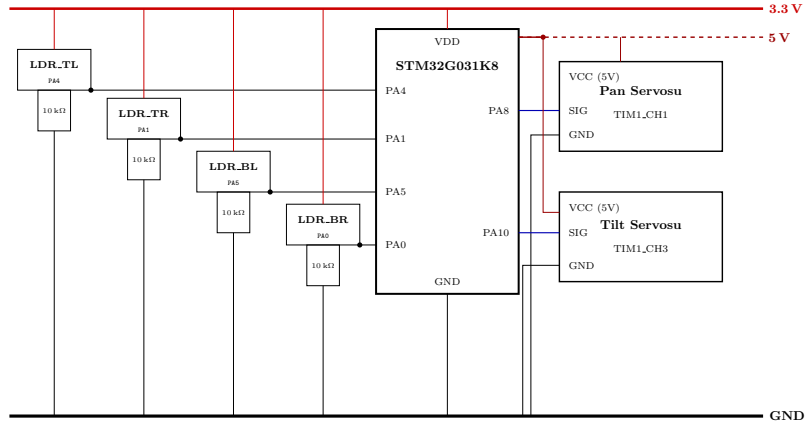


Figure 4.1: System circuit schematic (1/2)

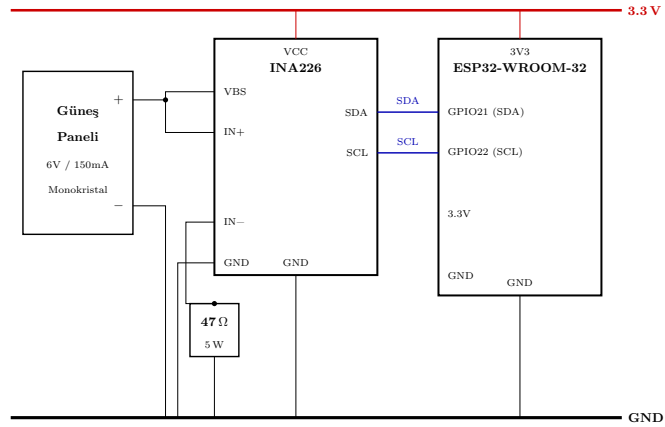


Figure 4.2: System circuit schematic (2/2)

5. Sensors

5.1 LDR (Light-Dependent Resistor)

An LDR is a photoconductive element whose resistance decreases proportionally with incident light intensity — from several $M\Omega$ in darkness to a few hundred Ω under full illumination.

Four LDRs are placed at the corners of the panel frame in a cross configuration: top-left, top-right, bottom-left, and bottom-right. Each sensor forms a voltage divider with a $10\text{ k}\Omega$ fixed resistor, with the output fed to an STM32 ADC input. The differential signal method is defined as:

$$e_{pan} = ADC_{left} - ADC_{right} \quad (5.1)$$

$$e_{tilt} = ADC_{top} - ADC_{bottom} \quad (5.2)$$

When $e_{pan} > 0$ the panel rotates left; when $e_{tilt} > 0$ it tilts upward. Both errors approaching zero indicates the panel is perpendicular to the sun.

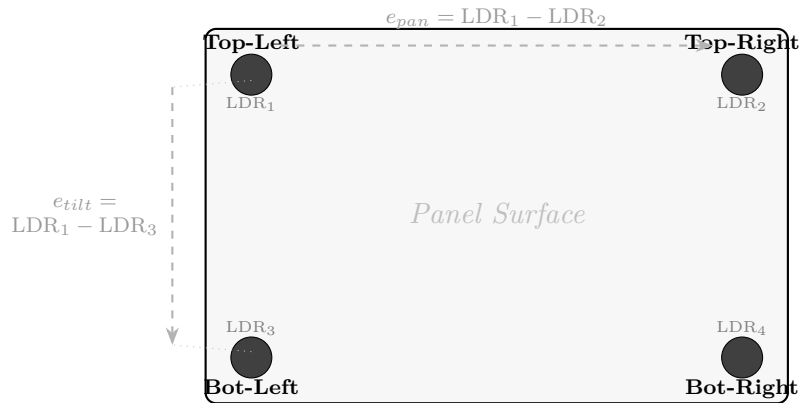


Figure 5.1: Cross-placement of LDR sensors at panel corners with differential signal definitions

5.2 INA226 Power Monitor

The INA226 is an I2C power monitor that measures the voltage drop across a shunt resistor to compute current, bus voltage, and power with high accuracy [4]. A $0.1\ \Omega$ shunt resistor is used with a maximum measurement range of 1.5 A .

Listing 5.1: INA226 initialisation and calibration

```

1 ina226.setResistorRange(0.1, 1.5);    // 0.1 Ohm shunt, max 1.5
   A
2 ina226.setAverage(INA226_AVERAGE_16); // 16-sample hardware
   averaging

```

Hardware averaging over 16 samples suppresses high-frequency noise at the sensor level. Cumulative energy is computed via discrete integration:

$$E_{Wh} = \sum P_W \cdot \frac{\Delta t}{3600} \quad (5.3)$$

6. Actuators

6.1 Servo Motors

Servo motors are angle-positioning actuators controlled by the duty cycle of a PWM signal. Typically, a 1 ms pulse corresponds to 0° and a 2 ms pulse to 180°.

Two servo motors are used — one for the pan (azimuth) axis and one for the tilt (elevation) axis — driven by STM32 Timer1 PWM outputs independently. Since the position value is stored as a `float`, a rounding correction is applied at the PWM write step to prevent quantisation jitter:

Listing 6.1: Float-preserving PWM write

```
1 __HAL_TIM_SET_COMPARE(&htim1, TIM_CHANNEL_1,
2                       (uint16_t)(pos_pan + 0.5f));
```

The `+0.5f` offset converts truncation to proper rounding, eliminating the last source of micro-stepping artifacts. A soft-start routine ramps the servos to the home position over 1 second on power-up.

7. Control Strategy

The system employs a six-layer signal processing pipeline in which each layer addresses a specific real-world problem. The full pipeline is illustrated in Figure 7.1 and all tunable parameters are summarised in Table 7.1.

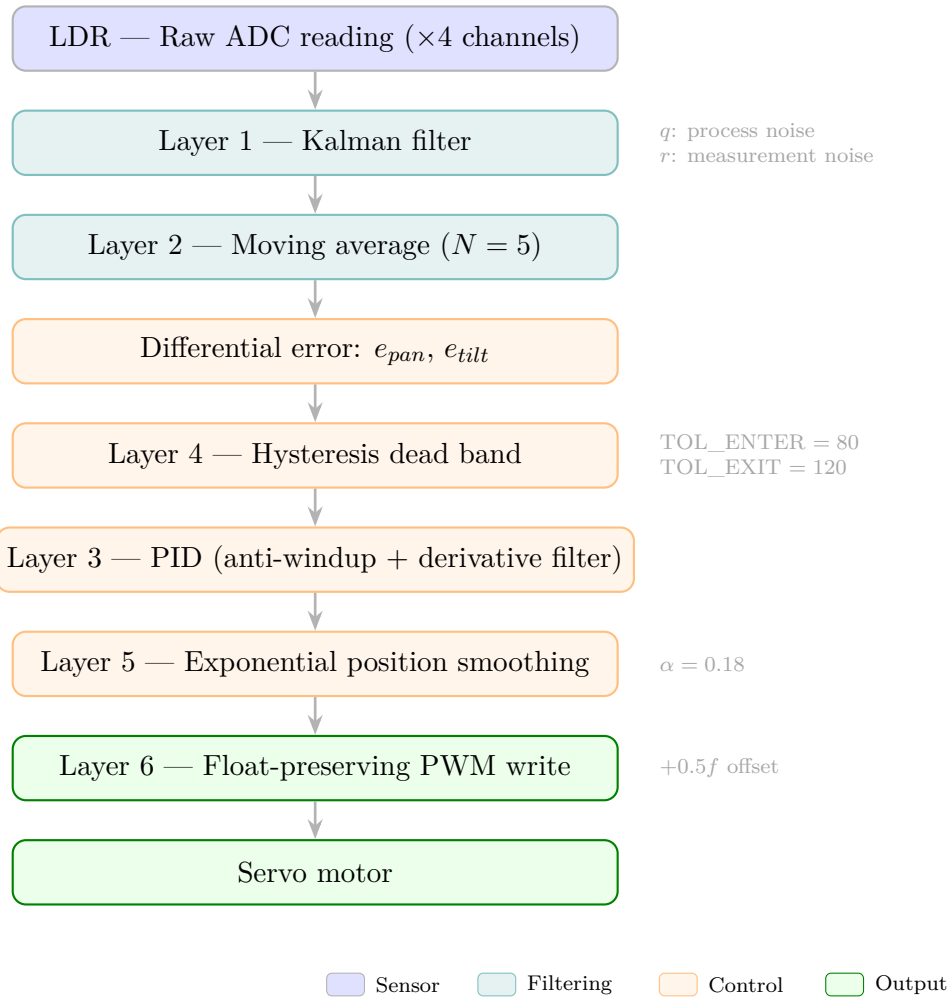


Figure 7.1: Control system signal processing pipeline

Table 7.1: Control and system parameter summary

Layer / Component	Parameter	Value
Kalman filter	Process noise q	Tunable
	Measurement noise r	Tunable
Moving average	Window size N	5
Hysteresis dead band	Activation threshold TOL_ENTER	80
	Deactivation threshold TOL_EXIT	120
Exponential smoothing	Smoothing factor α	0.18
PWM write	Rounding offset	$+0.5f$
INA226	Shunt resistance	$0.1\ \Omega$
	Maximum current	1.5 A
	Hardware averaging	16 samples
Solar panel	Rated voltage	6 V
	Rated current	150 mA
	Maximum power	0.9 W
Dummy load	Resistance	$47\ \Omega$
Servo (MG996R)	Idle consumption ($\times 2$)	1700 mW
	Active duty cycle	2.5%

7.1 Layer 1 — Kalman Filter

Raw ADC readings contain significant high-frequency noise, particularly under partially cloudy conditions. A 1D Kalman filter runs independently on each of the four LDR channels, separating genuine light-level changes from sensor noise [5].

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - \hat{x}_{k|k-1}) \quad (7.1)$$

Increasing the measurement noise parameter r produces a smoother but slower response.

7.2 Layer 2 — Moving Average ($N = 5$)

A 5-sample circular moving average is applied on top of the Kalman output. This dual-stage filtering virtually eliminates transient spikes caused by cloud edges or mechanical vibration.

7.3 Layer 3 — PID Controller

The differential error feeds into a full PID controller with two key enhancements:

Anti-windup (integral clamping): When the servo is at its mechanical limit, the integrator is hard-clamped to $\pm i_{limit}$ to prevent windup and overshoot on release.

Filtered derivative: A first-order low-pass filter is applied to the raw derivative term before it reaches the output:

$$d_{filtered} = \alpha \cdot d_{raw} + (1 - \alpha) \cdot d_{filtered,prev} \quad (7.2)$$

7.4 Layer 4 — Hysteresis Dead Band

A two-threshold hysteresis band prevents the chattering that occurs when using a single threshold:

- `TOL_ENTER = 80`: Motor activates when error exceeds this value
- `TOL_EXIT = 120`: Motor deactivates only when error drops below this value

7.5 Layer 5 — Exponential Position Smoothing

The target servo position is passed through a first-order exponential smoother to produce an organic, inertia-like motion profile:

$$pos = pos + \alpha \cdot (target - pos), \quad \alpha = 0.18 \quad (7.3)$$

7.6 Layer 6 — Float-Preserving PWM Write

Direct `float` $\rightarrow$ `uint16_t` casting truncates rather than rounds, creating invisible one-step quantisation jitter. The `+0.5f` offset corrects this.

8. Software Architecture

8.1 STM32 Firmware

The STM32 firmware is written in C using the HAL library. The main loop executes ADC reading, Kalman + moving average filtering, PID computation, and PWM write in sequence.

8.2 ESP32 Web Server

The ESP32 exposes the following HTTP endpoints via the AsyncWebServer library:

Table 8.1: ESP32 HTTP endpoints

Endpoint	Function
/	Live web dashboard (HTML)
/data	Real-time measurement data (JSON)
/start	Begin CSV recording session
/stop	Stop recording and prepare download
/csv	Download recorded data as CSV
/reset	Clear all runtime data

The last 3600 measurements (1 hour) are held in a circular buffer in RAM and served via chunked transfer, avoiding large heap allocations.

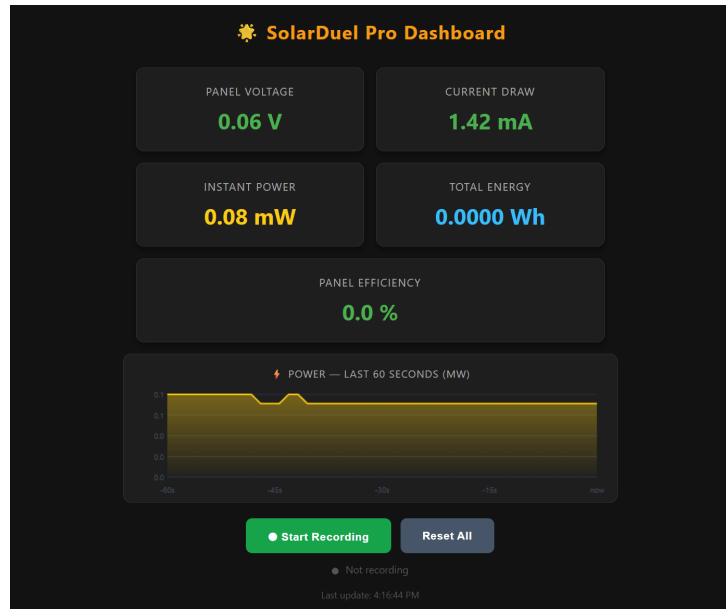


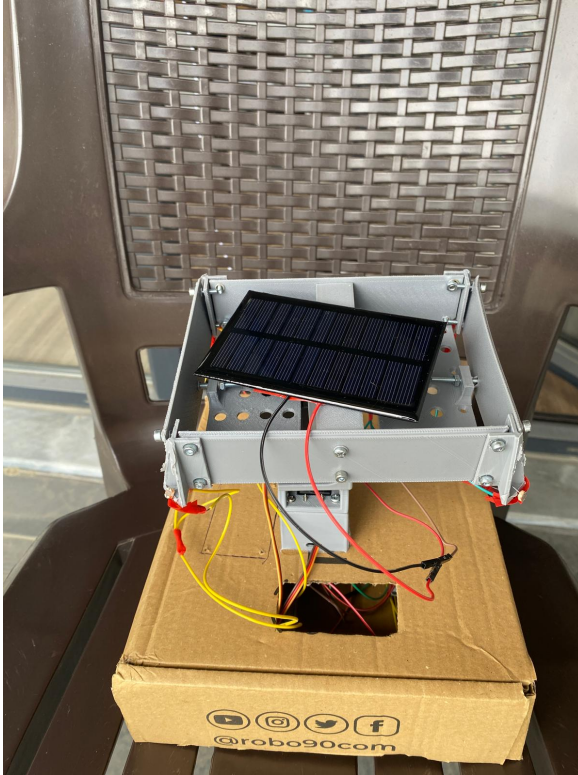
Figure 8.1: ESP32 live web dashboard

9. Data Collection and A/B Test Method

Since both panels cannot be connected to the INA226 simultaneously, an A/B test methodology was used. Under matched irradiance conditions, the tracker panel and the

fixed panel were each connected to the same dummy load in separate sessions. Recording was initiated via the *Start Recording* button on the dashboard and the resulting CSV was downloaded with *Stop & Download CSV*.

Each dataset contains time (s), voltage (V), current (mA), power (mW), and cumulative energy (Wh). The dataset used for analysis consists of 10 800 rows (3-hour session, 1 sample per second).



(a) Tracker panel session



(b) Fixed panel session

Figure 9.1: A/B test setup: each panel connected separately to the same load for matched measurements

10. Efficiency Analysis

The MATLAB analysis script (`efficiency_analysis.m`) processes both CSV files and produces a four-panel comparative figure.

10.1 Results Summary

Table 10.1: Tracker vs fixed panel comparative efficiency results

Metric	Tracker	Fixed Panel
Peak Efficiency	74.44%	35.00%
Average Efficiency	52.31%	21.14%
Peak Power	670.0 mW	315.0 mW
Average Power	470.8 mW	190.2 mW
Total Energy Harvested	1.4125 Wh	0.5707 Wh
Energy Gain	+147.5%	—
Avg. Instantaneous Gain	180.3%	—

10.2 Analysis Figures

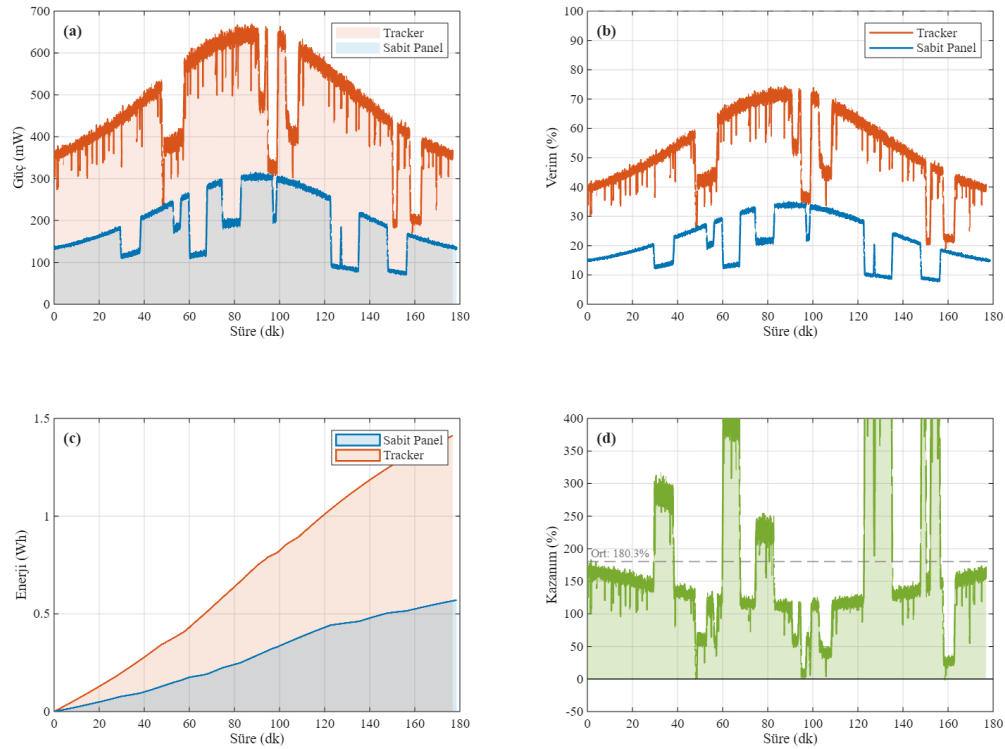


Figure 10.1: Comparative efficiency analysis: (a) instantaneous power, (b) panel efficiency, (c) cumulative energy, (d) tracker gain over fixed

Figure 10.1(a) shows that the tracker consistently delivers higher power throughout the session, with characteristic cloud-induced dips visible in both curves. Figure 10.1(c) shows the tracker harvesting 1.41 Wh versus 0.57 Wh for the fixed panel over the same period. The average instantaneous gain of 180.3% shown in Figure 10.1(d) confirms the advantage of continuous sun-facing orientation.

10.3 Net Energy Gain and Parasitic Consumption Analysis

For the gross energy advantage to be meaningful from an engineering standpoint, the parasitic energy consumed by the tracking system itself must be accounted for. This analysis is critical for evaluating the true efficiency of solar tracking systems, as emphasised by Seme et al. [2].

10.3.1 Parasitic Consumption Estimate

Table 10.2: Tracker unit component power consumption

Component	Idle (mW)	Active (mW)	Source
STM32G031K8	15	15	Datasheet
Servo Motor $\times 2$ (MG996R)	1700	5000	MG996R Datasheet
Total	1715	5015	

The servos are not continuously active; they engage only when the LDR differential error exceeds the hysteresis threshold (`TOL_ENTER = 80`). Given that the sun moves approximately 15° per hour, the servo active duty cycle is estimated at approximately 2.5% (roughly 3 seconds of motion every 2 minutes).

10.3.2 Net Energy Calculation

Parasitic energy over the 3-hour session:

$$E_{STM32} = 15 \text{ mW} \times 3 \text{ h} = 45 \text{ mWh} \quad (10.1)$$

$$E_{servo, idle} = 1700 \text{ mW} \times (1 - 0.025) \times 3 \text{ h} = 4972 \text{ mWh} \quad (10.2)$$

$$E_{servo, active} = 5000 \text{ mW} \times 0.025 \times 3 \text{ h} = 375 \text{ mWh} \quad (10.3)$$

$$E_{parasitic} = 45 + 4972 + 375 = 5392 \text{ mWh} \approx 5.392 \text{ Wh} \quad (10.4)$$

Table 10.3: Net energy gain analysis (3-hour session)

Parameter	Value (Wh)	Note
Tracker gross energy	1.4125	INA226 measurement
Fixed panel energy	0.5707	INA226 measurement
Gross energy gain	+0.8418	Tracker – Fixed
Parasitic consumption	–5.3920	STM32 + $2 \times$ MG996R
Net energy gain	–4.550	Gross – Parasitic

The net energy analysis reveals that the MG996R servos present a disproportionate parasitic load for a 6 V/150 mA panel: each servo draws approximately 850 mW at idle, close to the panel’s entire rated output of 900 mW. This finding underscores the critical

importance of actuator selection in small-scale tracking systems. A redesign using lower-power micro-servos (e.g., SG90: ~ 50 mW/servo idle) or higher-capacity panels is expected to yield a positive net energy gain.

11. Conclusions

The dual-axis solar tracking system presented in this work produces 147.5% more gross energy than the fixed panel configuration over a three-hour measurement session. The net energy analysis shows that the MG996R servo motors consume more power at idle than the panel produces, highlighting that actuator selection is the dominant factor in the energy balance of small-scale tracking systems.

The multi-layer control architecture effectively suppresses high-frequency LDR noise and servo chattering. The physical separation of the STM32 tracker unit and the ESP32 logger unit ensures that data acquisition is not affected by motor control latency.

Strengths of the system include the real-time web dashboard, browser-side CSV recording, and hardware-level measurement averaging (INA226 16-sample average). Limitations include the sequential A/B test methodology, which cannot capture both panels simultaneously, and the 3-hour measurement window.

Bibliography

- [1] H. Mousazadeh, A. Keyhani, A. Javadi, H. Mobli, K. Abrinia, and A. Sharifi, “A review of principle and sun-tracking methods for maximizing solar systems output,” *Renewable and Sustainable Energy Reviews*, vol. 13, no. 8, pp. 1800–1818, 2009.
- [2] S. Seme, G. Stumberger, and J. Vorsic, “Maximum efficiency trajectories of a two-axis sun tracking system determined considering tracking system consumption,” *IEEE Transactions on Power Electronics*, vol. 26, no. 4, pp. 1280–1290, 2011.
- [3] Instructables, “Solar tracker tilt/pan panel frame ldr mounts rig,” 2020.
- [4] Texas Instruments, *INA226 Bi-Directional Current/Power Monitor Datasheet*, 2015.
- [5] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

A. STM32 Control Loop — Core Flow

Listing A.1: STM32 main control loop (simplified)

```

1  while (1) {
2      // 1. Read ADC
3      adc_top    = HAL_ADC_Read(CH1);
4      adc_bot    = HAL_ADC_Read(CH2);
5      adc_left   = HAL_ADC_Read(CH3);
6      adc_right  = HAL_ADC_Read(CH4);
7
8      // 2. Kalman + moving average
9      kalman_update(&kf_top, adc_top);
10     ma_top = moving_avg(kf_top.x);
11     // ...
12
13     // 3. Differential error
14     e_pan  = ma_left - ma_right;
15     e_tilt = ma_top  - ma_bot;
16
17     // 4. Hysteresis + PID + exponential smoothing + PWM write
18     if (fabs(e_pan) > TOL_ENTER) {
19         pid_out_pan = pid_compute(&pid_pan, e_pan);
20         pos_pan = pos_pan + 0.18f * (pid_out_pan - pos_pan);
21         __HAL_TIM_SET_COMPARE(&htim1, TIM_CHANNEL_1,
22                               (uint16_t)(pos_pan + 0.5f));
23     }
24     HAL_Delay(20);
25 }
```